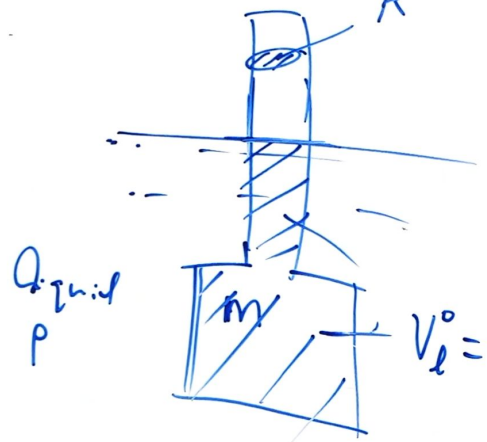
30-6-26

Waves Lec 6

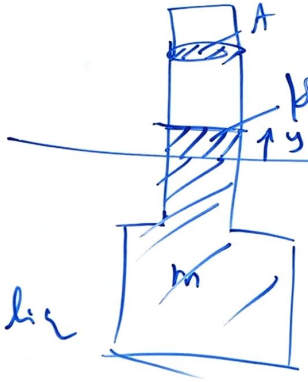
Tutorial – 1 A.P. French, Vibrations & Waves
ch. 1, Probs 1-10, 11, 12, ch. 3, 3-1,



ρ_{liquid} = ρ

V_l° = equil vol. of liq displaced

m_l° = equil mass of liq displaced
$= \rho V_l^\circ$



$V_l = V_l^\circ - Ay$

$m_l = \rho V_l$
$= \rho (V_l^\circ - Ay)$

$m \frac{d^2y}{dt^2} = -mg + m_l g$
$= (m_l - m) g$

In Equil.

$y = 0, v = 0$

$\frac{d^2y}{dt^2} = 0 \Rightarrow F_z^{net} = 0$

$y = 0, m_l = m_l^\circ$

$0 = (m_l^\circ - m) g$

$\Rightarrow m_l^\circ = m$

$\downarrow$

$\rho V_l^\circ = m$

$V_l^\circ = m/\rho$

$m_l = \rho V_l^\circ - \rho A y$
$= m - \rho A y$

$m_l - m = (m - \rho A y) - m$

$= -\rho A y$

$m \frac{d^2y}{dt^2} = (m_l - m)\, g$

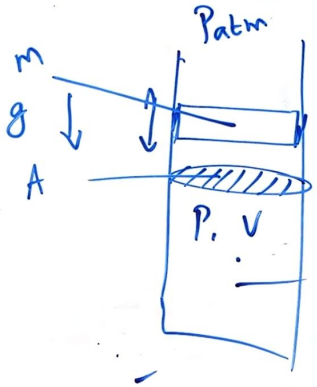
$= -\rho g A y$

$= -k y$

$k = \rho g A$

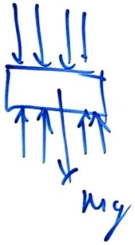
$\omega^2 = \frac{k}{m} = \frac{\rho g A}{m}$

$\Delta(xy) = x\Delta y + y\Delta x$

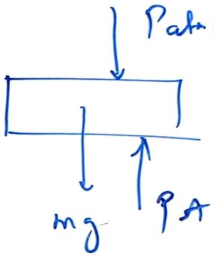


$P = P_{atm} + \frac{mg}{A}$

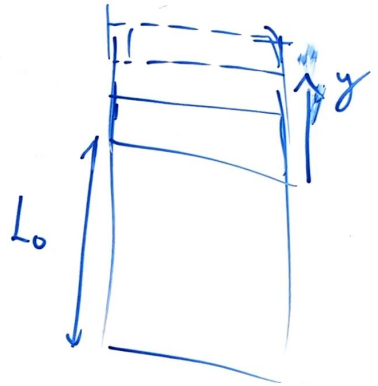
in equil.



$\frac{k}{m} = \frac{P_0 A}{m L_0}$



$PV = nRT$



$m \frac{d^2y}{dt^2} = -\frac{P_0 A y}{L_0}$

$F = -\frac{P_0 A y}{L_0}$

$T = \text{const}$

$PV = \text{const}$

$\Delta V = A y$

$P\Delta V + V\Delta P = 0$

$\Delta P = -P\frac{\Delta V}{V}$

$\Delta P\big|_{V=V_0} = -P\frac{Ay}{V}\Big|_{V=V_0}$

$= -P_0 \frac{Ay}{V_0}$

$F = \Delta P A$

$V_0 = A L_0$